

Data Science Across Domains

From Methods and Models to
Business Applications



Rodrigo Kang

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Welcome

Welcome to my portfolio:

Data Science Across Domains

From Methods and Models to Business Applications

This portfolio presents a curated collection of applied data science projects developed across different domains. Rather than a traditional theoretical text, it is organized as a series of case studies showing how modern data science methods are used to frame problems, build models, and generate practical insights.

Each chapter focuses on a specific project. Problems are clearly defined, modeling assumptions are made explicit, and solutions are developed using a combination of statistical reasoning, machine learning methods, and computational experimentation. Results are evaluated using appropriate metrics and baseline comparisons.

The emphasis throughout the portfolio is on applied problem solving. Mathematical concepts and technical details are introduced only when they help explain the modeling choices or the behavior of the algorithms being used.

Every project presented here corresponds to a fully implemented workflow. The associated computational artifacts—written in Python and R—are available alongside the text, allowing readers to explore the code, reproduce the results, and understand the implementation details behind each solution.

This portfolio reflects how data science is practiced in real environments: combining sound modeling principles, careful experimentation, and clear communication of results.

Many of the projects presented here also draw on my background in physics and quantitative modeling, where mathematical reasoning and computational

experimentation naturally complement modern machine learning techniques.

Each chapter can be read independently as a standalone project, while together they illustrate a consistent approach to building and applying data science solutions across different domains.

Preface

This portfolio reflects an ongoing effort to organize my work in applied data science into a coherent and practice-oriented body of projects. Rather than presenting isolated experiments or purely theoretical discussions, it documents complete workflows in which problem formulation, modeling choices, implementation, and evaluation are treated as parts of the same analytical process.

Structuring this material in book form mirrors how I approach machine learning problems in professional and research environments. In practice, successful applications rarely depend on algorithmic novelty alone. They usually emerge from a clear definition of the problem, explicit modeling assumptions, careful experimentation, and thoughtful interpretation of results within the context of the domain.

For this reason, the chapters are organized around concrete case studies. Each project begins with a well-defined problem and progresses through model design, training strategy, evaluation methodology, and discussion of limitations. Concepts from statistics, optimization, probability, and numerical methods appear throughout the text, but only when they are necessary to support modeling decisions or clarify the behavior of the methods being used.

An equally important aspect of this portfolio is its deliberate scope. It does not attempt to catalogue every possible technique or follow trends for their own sake. Instead, it focuses on methods that I have implemented, tested, and evaluated in real problem settings. The emphasis is on depth, clarity, and technical responsibility rather than breadth.

All projects presented here correspond to fully implemented solutions. Computational artifacts written in Python and R accompany the text and are referenced throughout, allowing readers to explore the implementation and reproduce the

results if desired.

Ultimately, this portfolio serves both as a record of completed work and as a foundation for continued development. The projects documented here represent practical steps in an ongoing process of learning, experimentation, and refinement. My intention is that the structure of this work reflects not only technical capability, but also a disciplined and pragmatic approach to applying data science methods in real-world contexts.

About

I am a physicist and data scientist with a background in applied mathematics, machine learning, and computational modeling. My work focuses on applying quantitative methods to real-world problems, combining mathematical reasoning with practical implementation.

My academic training in physics has strongly shaped the way I approach data science and machine learning. My research included work on **quantum optimal control theory**, with particular emphasis on algorithmic design based on *shortcuts to adiabaticity* and the study of computational complexity in physical systems. This background continues to influence my modeling philosophy: clear problem formulation, explicit assumptions, and careful evaluation of model behavior.

Professionally, I have applied machine learning, advanced analytics, and optimization techniques in data-intensive industries. Much of my work has been developed within the **energy sector**, where I have contributed to quantitative modeling, decision-support systems, and optimization frameworks for gas trading and operational planning.

In the retail sector, I have worked on projects involving customer segmentation, churn prediction, and recommendation systems, balancing theoretical rigor with practical constraints such as data quality, interpretability, and deployment considerations.

More recently, within Oil & Gas research and development environments, my work has focused on integrating physical modeling, statistical inference, and machine learning to address complex industrial problems. These projects include predictive modeling for pipeline corrosion, machine learning approaches to production optimization, and simulation-based frameworks for decision-making un-

der uncertainty.

Across both academic and industrial settings, my perspective remains consistent: machine learning methods such as Deep Learning and Reinforcement Learning are powerful computational tools, but their effectiveness ultimately depends on disciplined modeling, transparent assumptions, and careful validation.

Outside my professional work, I enjoy playing the bass guitar and maintain a longstanding interest in theology.

How to Read This Portfolio

This portfolio is organized around applied projects rather than purely theoretical chapters. Each chapter presents a self-contained case study illustrating how data science methods can be used to address a specific problem, from initial formulation and modeling to evaluation and interpretation.

Projects can be read independently. Readers interested in a particular topic can navigate directly to the corresponding chapter without needing to follow the entire sequence.

Most projects follow a common structure:

1. **Problem definition**
A clear description of the objective and the context in which the problem arises.
2. **Modeling approach**
Selection and development of appropriate statistical or machine learning methods.
3. **Implementation**
Practical aspects of the solution, including data preparation, model training, and computational considerations.
4. **Evaluation**
Performance assessment using appropriate metrics and baseline comparisons.
5. **Discussion and limitations**
Interpretation of results, assumptions behind the model, and potential areas for improvement.

Theoretical concepts appear only when they help clarify modeling decisions or explain the behavior of the methods being used. The emphasis throughout this work is on applied problem solving rather than abstract presentations of algorithms.

Each chapter is accompanied by computational artifacts implemented in Python and R, allowing readers to explore the code and reproduce the results if desired.

The material presented here combines quantitative reasoning, machine learning methods, and computational experimentation, with a consistent focus on building solutions that are both technically sound and practically meaningful across different domains.

Part I

Quantitative Finance

Overview

This part covers problems related to quantitative finance, including the modeling of financial assets, portfolio construction, and risk analysis under uncertainty.

The scope of this domain includes methods for analyzing market data, evaluating financial instruments, and supporting decision-making where risk and return must be jointly considered.

Chapter 1

CAPM and Fama–French Models

This project studies how classical and multi-factor asset pricing models explain the relationship between systematic risk and expected return. Using historical market data, the analysis evaluates the assumptions, explanatory power, and empirical limitations of the CAPM and Fama–French frameworks through regression-based factor modeling.

1.1 Problem Definition

This project evaluates how asset pricing models—from the traditional CAPM to the multi-factor extensions of Fama and French—formalise the relationship between risk and expected return. In practice, an investor faced with a price signal must determine whether the potential return justifies the associated risk. Doing this effectively means discounting uncertain future cash flows back to present value, which requires a reliable discount rate. The core question is straightforward: what minimum return should be demanded from an asset given its specific risk profile?

Getting this right is critical for asset valuation, portfolio construction, and fund performance evaluation. Without a structured framework, investment decisions often default to guesswork or oversimplified rules of thumb—such as chasing past returns or assuming that any form of risk automatically deserves a premium. Modern financial theory offers a more rigorous perspective: the market does not reward idiosyncratic or stock-specific risk, since it can be diversified away.

Only systematic risk warrants a higher expected return. This principle is what separates the CAPM from more naive approaches and serves as the foundation for multi-factor models.

Key decisions—ranging from efficient capital allocation to identifying market mispricings—rely heavily on these models. If an asset offers an expected return above what its systematic risk predicts, it may indicate undervaluation. However, the real-world challenge is methodological: true expected returns are unobservable. Relying on historical data introduces noise, sample-period sensitivity, and specification errors. Furthermore, the CAPM compresses all systematic risk into a single factor (the market portfolio), a simplification that empirical evidence has repeatedly challenged.

The Fama–French models attempt to address this limitation by capturing additional dimensions of systematic risk, including size, value, profitability, and investment patterns. From a data science perspective, this project explores how these models are constructed, tests their underlying assumptions, and evaluates how well they fit empirical data. The final output is a comparative analysis of the models’ explanatory power and practical limitations when confronted with real-world financial data that rarely behaves as textbook theory predicts.

1.2 Modeling Approach

To approach the problem of asset valuation, this project is grounded on two fundamental principles of finance. The first is the time value of money: a dollar today is worth more than a dollar tomorrow because it can be invested immediately and start generating returns. The second is risk aversion: an uncertain cash flow is only attractive if it offers additional compensation relative to a risk-free alternative. Together, these principles reflect the two central variables underlying any financial valuation problem: time and risk.

Under this framework, the valuation of an asset can be understood as the present value of an uncertain future cash flow:

$$PV = \left(\frac{1}{1 + r_t} \right)^t \mathbb{E}[X_t] \quad (1.1)$$

Here, PV represents the expected present value of a future cash flow X_t received

at time t , while r_t denotes the discount rate associated with risk and the opportunity cost of capital. The fundamental challenge for the investor is that this discount rate is not known in advance. In other words, the key question is how much return should be required to compensate for the level of risk being taken. This problem provides the conceptual motivation for the Capital Asset Pricing Model (CAPM) and its multifactor extensions.

Since future returns are uncertain, this project models them as random variables. A practical way to characterize them is through their expected value, representing the anticipated average return, and their volatility, which measures the dispersion of outcomes around that average. This mean-variance framework, originally developed by Harry Markowitz in modern portfolio theory, shows that total portfolio risk can be reduced by combining assets whose returns do not move exactly in the same way [1]. The intuition is straightforward: through diversification, part of the individual fluctuations of each asset tends to offset one another.

However, diversification has an observable practical limit. As portfolios are constructed with an increasing number of assets, total volatility initially falls rapidly but eventually stabilizes around a lower bound. As illustrated in Figure 1.1, this behavior reveals the existence of two distinct components of risk. On the one hand, idiosyncratic or diversifiable risk, which is associated with firm-specific events and can largely be eliminated through diversification. On the other hand, systematic risk, which is linked to broader macroeconomic and market-wide movements and therefore persists even in highly diversified portfolios.

This distinction has a fundamental implication for asset pricing. In a competitive market, rational investors should not expect compensation for taking risks that can be eliminated at virtually no cost through diversification. As a result, the market only rewards exposure to systematic risk. This idea forms the conceptual foundation of the CAPM, which states that the expected risk premium of an asset should be proportional to the expected risk premium of the market [2] [3]:

$$\mathbb{E}[r_i] - r_f = \beta_i (\mathbb{E}[r_m] - r_f) \quad (1.2)$$

In this expression, $\mathbb{E}[r_i]$ represents the expected return of asset i , r_f is the risk-free rate, and $\mathbb{E}[r_m]$ corresponds to the expected return of the market portfolio. The parameter β_i measures the sensitivity of the asset to market fluctuations and summarizes its exposure to systematic risk.

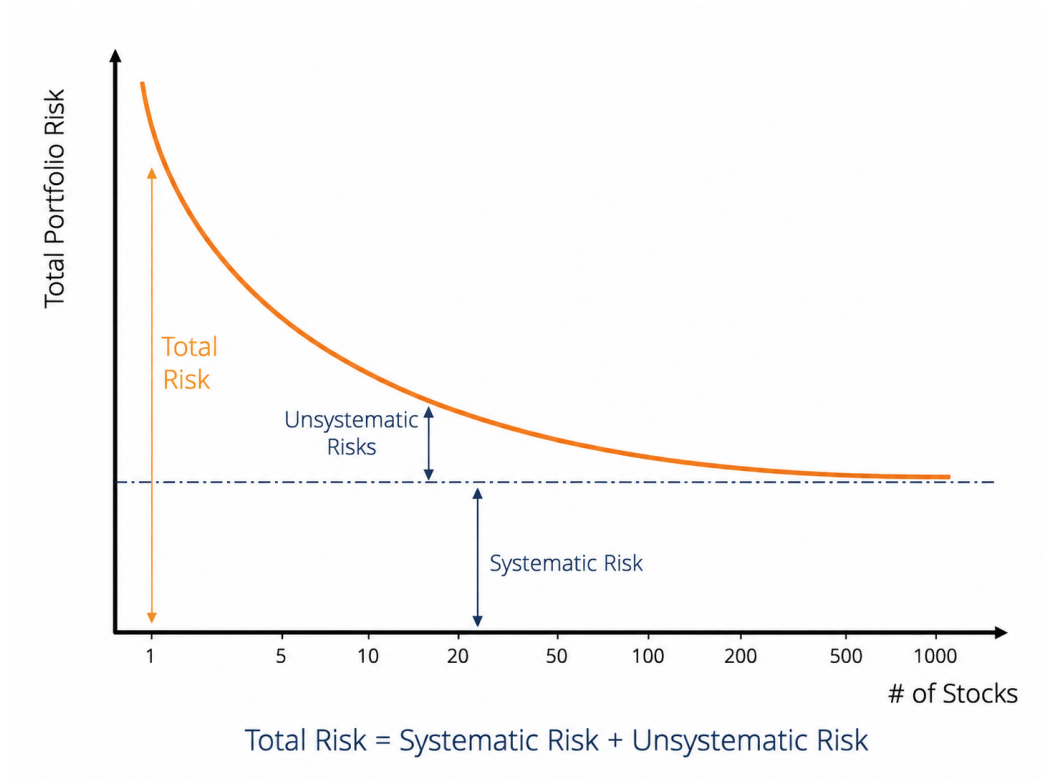


Figure 1.1: Decomposition of total risk into systematic and diversifiable risk as assets are progressively added to a portfolio.

From a modeling perspective, the estimation of beta can be interpreted as a simple linear regression between asset returns and market returns [4]. In this project, the CAPM is implemented precisely under this framework, using simple linear regression to estimate the market exposure of different assets and analyze the relationship between systematic risk and expected return.

Despite its conceptual simplicity and historical importance, the CAPM assumes that the entire structure of systematic risk can be summarized by a single factor: the market itself. Empirical evidence, however, suggests that certain firm characteristics — such as company size or the relationship between book value and market value — capture persistent return patterns that the CAPM does not fully explain. These observations motivated the development of the multifactor models proposed by Fama and French [5].

In this project, these extensions are implemented using multiple linear regression. The three-factor model can be expressed as:

$$\mathbb{E}[r_i] - r_f = \alpha + \beta_{mkt}(\mathbb{E}[r_m] - r_f) + \beta_{smb} \cdot \text{SMB} + \beta_{hml} \cdot \text{HML} + \varepsilon \quad (1.3)$$

where SMB (*Small Minus Big*) measures the excess return of small-cap stocks relative to large-cap stocks, while HML (*High Minus Low*) captures the return differential between value and growth stocks [6] [7].

From a modeling perspective, the transition from the CAPM to the Fama–French three-factor model can be interpreted as moving from a simple linear regression to a multiple linear regression framework, incorporating additional explanatory variables to capture dimensions of systematic risk that a single-factor model cannot fully represent.

The main objective of the project is not simply to compare statistical goodness-of-fit metrics, but to understand how different economic factors contribute to the generation of returns in financial markets and to evaluate how effectively models of different complexity levels represent that underlying process.

1.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

1.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

1.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches

- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

1.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

1.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 2

Asset Allocation with PCA and Hierarchical Clustering

Work in progress.

2.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

2.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

2.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
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Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

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Typical elements may include:

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- Error analysis
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- Potential future improvements

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This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Part II

Retail Analytics

Overview

This part covers problems related to retail analytics, including the analysis of customer behavior, transaction data, and marketing performance.

The scope of this domain includes methods for segmentation, prediction, and decision support, with applications in areas such as fraud detection, customer lifetime value estimation, and campaign optimization.

Chapter 3

Credit Card Fraud Detection

Work in progress.

3.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

3.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

3.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

3.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

3.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

3.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

3.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 4

Customer Lifetime Value Prediction

Work in progress.

4.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

4.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

4.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

4.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

4.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

4.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

4.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 5

Marketing Campaign Optimization

Work in progress.

5.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

5.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

5.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

5.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

5.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

5.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

5.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

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The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 6

RFM Segmentation with K-Means

Work in progress.

6.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

6.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

6.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

6.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

6.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

6.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
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6.7 Technical Appendix (Optional)

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- Extended equations
- Supplementary experiments

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Chapter 7

Customer Sentiment and Topic Analysis

Work in progress.

7.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

7.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

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7.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

7.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

7.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

7.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

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7.7 Technical Appendix (Optional)

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Possible contents include:

- Mathematical derivations
- Additional proofs
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- Extended equations
- Supplementary experiments

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Part III

Biological Systems

Overview

This part covers problems related to biological systems, including the analysis and modeling of data arising in biological, health, and life science contexts.

The scope of this domain includes approaches for studying complex processes, identifying patterns in biological data, and supporting analysis in settings where systems are often heterogeneous, nonlinear, and influenced by multiple interacting factors.

Chapter 8

Microbial Population Dynamics (Lotka–Volterra Model)

Work in progress.

8.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

8.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

8.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

8.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

8.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

8.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

8.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
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- Supplementary experiments

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Part IV

Energy and Industrial Systems

Overview

This part covers problems related to energy and industrial systems, including the analysis and modeling of production processes, physical assets, and operational data.

The scope of this domain includes approaches for optimization and efficient resource allocation, monitoring and analysis of system integrity, and the study of degradation processes such as corrosion, as well as data-driven methods to support decision-making in complex operational environments.

Chapter 9

Geochemical Fingerprint Deconvolution

Work in progress.

9.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

9.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

9.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

9.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

9.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

9.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
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- Trade-offs associated with the methodology
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Possible contents include:

- Mathematical derivations
- Additional proofs
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- Supplementary experiments

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Chapter 10

Pore–Throat Detection in Porous Media

Work in progress.

10.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

10.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

10.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

10.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

10.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

10.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
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Chapter 11

Optimal Hedging of Industrial Gas Demand

Work in progress.

11.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

11.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

11.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

11.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures

- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

11.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

11.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
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- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

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Possible contents include:

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Chapter 12

Knowledge Extraction from Technical Documents with NLP and LLMs

Work in progress.

12.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

12.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

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The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

12.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

12.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization

- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

12.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
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- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

12.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
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- Sources of uncertainty
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12.7 Technical Appendix (Optional)

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Part V

Climate and Environmental Systems

Overview

This part covers problems related to climate and environmental systems, including the analysis and modeling of temporal and spatial data describing atmospheric and large-scale natural processes.

The scope of this domain includes approaches for time series analysis, forecasting, and the study of variability and uncertainty in systems influenced by natural dynamics, external drivers, and interactions across multiple scales.

Chapter 13

Climate Forecasting with Time Series

Work in progress.

13.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

13.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

13.3 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

13.4 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

13.5 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
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- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

13.6 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
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This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

13.7 Technical Appendix (Optional)

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Possible contents include:

- Mathematical derivations
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Part VI

Writings & Essays

Overview

This part covers technical writings and essays spanning scientific modeling, data analysis, and related areas of mathematics, physics, and philosophy.

The scope includes discussions of modeling principles, reasoning under uncertainty, and computational approaches, providing context and deeper insight into the ideas underlying the projects.

Chapter 14

Data Science in the Age of Hype: Why the scientific method still matters

14.1 The Rise of Data Science

Data has become one of the most valuable assets of any modern organization. The unprecedented flow of information—resulting from countless digital interactions such as clicks, e-commerce transactions, mobile communications, and social media activity—reflects its central role in contemporary society. At the same time, the rapid advances in computing that enable large-scale data collection, storage, and processing are radically transforming the way businesses, governments, scientific communities, and institutions make decisions.

Yet, raw data by itself is meaningless. It only acquires value when subjected to systematic and rigorous analysis through methodologies capable of detecting patterns, inferring relationships, predicting trends, and ultimately translating insights into decisions with real-world impact. This is why Clive Humby’s now-famous analogy describes data as “the new oil”: like crude oil, data must be refined before it can generate value. The crucial element is not the “raw material” itself but the techniques used to extract its potential and the knowledge derived from it. This reflection naturally leads to a fundamental question: what do we really mean by Data Science?

Historically, the term Data Science emerged from an evolutionary process shaped by multiple contributions that helped formalize it as a discipline. Although its roots lie in statistics and computer science, its consolidation as an autonomous field took place over more than half a century of theoretical and technological developments.

The conceptual seed of Data Science can be traced back to 1962, when American statistician John W. Tukey published his influential essay *The Future of Data Analysis*. In it, Tukey argued that statistics should not be confined to hypothesis confirmation alone, but should place equal emphasis on the exploration, diagnosis, and understanding of data [8], [9]. He introduced *data analysis* as a distinct discipline, anticipating the vision of a “science of learning from data”—one in which empirical inquiry and iterative interaction with data occupy the core of quantitative knowledge.

Later, in 1974, Danish computer scientist Peter Naur, in his *Concise Survey of Computer Methods*, explicitly used the terms *Data Science* and *Datalogy* to refer to the systematic study of data [10]. Naur argued that computing should focus on the structure, meaning, and organization of data rather than on hardware or programming. Although his proposal went largely unnoticed for years, it laid conceptual foundations that would be revisited decades later.

In 1997, statistician C.F. Jeff Wu, during his inaugural lecture at the University of Michigan titled *Statistics = Data Science?*, proposed renaming statistics as *Data Science* and calling modern statisticians *data scientists* [11]. His goal was to signal a new era in which statistical inference should integrate with computation and empirical analysis to move beyond mere data description and outdated stereotypes.

This perspective was further developed in 2001 by William S. Cleveland in his seminal article *Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics* [12]. Cleveland proposed a systematic expansion of statistics by identifying six core technical areas of Data Science: multidisciplinary investigations, models and methods for data, computing with data, pedagogy, tool evaluation, and theory. Rather than allowing statistics to contract into a narrowly mathematical discipline, he emphasized the central role of computation, real-world data analysis, education, and the development and evaluation of tools. His framework laid the conceptual groundwork for understanding Data Science as a comprehensive, data-centered field—deeply rooted in statistics, yet necessarily broader in scope and practice.

That same year, Leo Breiman, a statistician at the University of California, Berkeley, published his influential essay *Statistical Modeling: The Two Cultures*, in which he argued that statistical modeling serves two fundamental purposes: **information**—extracting insight into how nature associates response variables with input variables—and **prediction**—the ability to accurately anticipate future responses for new observations [13]. Breiman critically observed that traditional statistical practice had largely privileged the informational goal through highly structured, assumption-driven models, often at the expense of predictive performance. He advocated for a broader methodological perspective in which algorithmic and data-driven approaches play a central role, emphasizing prediction as an essential complement rather than a secondary objective. This distinction, and the tension it revealed between explanation and prediction, foreshadowed core principles of modern Data Science, where understanding and predictive accuracy are treated as intertwined but distinct aims.

By the late 2000s, the rise of Big Data and the growing need for professionals capable of combining statistics, programming, and domain expertise fueled the popularization of both the term Data Science and the role of the data scientist. Media consolidation came in 2012, when Thomas H. Davenport and DJ Patil published *Data Scientist: The Sexiest Job of the 21st Century* in the *Harvard Business Review* [14]. Since then, Data Science has become a strategic discipline across both academia and industry.

Today, Data Science is recognized as an interdisciplinary field that integrates statistical, computational, and conceptual methods to extract knowledge, infer patterns, and generate predictions from data. In many ways, it represents the natural evolution of statistics a discipline now equipped to address complex phenomena in fields as diverse as economics, biology, marketing, and physics.

However, despite its relevance and expansion, the term *Data Science* has become somewhat diffuse, often distorted by media overexposure. This ambiguity calls for a deeper examination of its epistemological foundations: if we speak of a **Science** of Data, we must ask what makes a discipline a science?

From this perspective, Data Science can indeed be considered a science. Its scientific character lies in the systematic application of the scientific method to the study of data. The data scientist formulates hypotheses, analyzes observations, and tests results just as any other scientist does. The difference lies in the object of study: data as a quantitative representation of reality.

In this sense, Data Science can be viewed as a generalization of statistics extending its focus from **inference** (understanding how the world behaves based on observed data) to **prediction** (anticipating how it might behave under new conditions). This predictive capability, powered by machine learning algorithms, enables the effective application of the **Hypothetical-Deductive Method**: formulating hypotheses about future phenomena and testing them through data and predictive models. In other words, machine learning acts as a technological enabler that materializes Breiman's inference–prediction duality, providing tools to model complex relationships directly from data and to test hypotheses that were previously unreachable with traditional statistical methods. It is this integration of inference, prediction, and empirical validation that solidifies the scientific nature of the discipline.

Understanding these foundations is essential for interpreting the true scope of Data Science. While the discipline draws from statistics, computer science, and domain expertise, its ultimate objective is not merely the manipulation of data but the extraction of reliable knowledge about the world. In this sense, Data Science should not be understood as a collection of algorithms or software tools, but as a methodological framework for reasoning under uncertainty.

However, the rapid growth of the field has also brought an unintended consequence: a proliferation of narratives that portray Data Science primarily as a technological revolution driven by ever more sophisticated algorithms. In many contexts, the emphasis has shifted toward tools, automation, and predictive performance, sometimes overshadowing the epistemological principles that historically guided scientific inquiry.

This tension raises a critical question for modern practitioners: if Data Science claims to be a science, what role does the scientific method play in its practice today? Addressing this question requires revisiting the principles that define scientific reasoning and examining how they apply to the analysis and interpretation of data in an era increasingly shaped by computational power and algorithmic models.

14.2 The Nature of Science and the Scientific Method

The very semantics of the term *Data Science* make it impossible to detach it from the concept of *science*. This forces us to pause and reflect: can Data Science truly

be regarded as a scientific discipline, or is it merely a fashionable commercial label? A comprehensive analysis of the nature of science lies far beyond the scope of this article; however, for the sake of intellectual honesty, we should not evade the issue. At the very least, we ought to clarify what makes an activity genuinely *scientific*.

From an epistemological perspective, science cannot be defined merely as any endeavor that leads to new discoveries or expands knowledge. Scientific knowledge evolves what was once considered true has often been revised, refined, or discarded. It is therefore reasonable to assume that part of what we regard today as scientifically valid will eventually be replaced, even if we cannot foresee which part. In other words, science does not guarantee truth. What distinguishes science from other forms of inquiry is not the *results* it produces, but the *method* it employs to generate them [15]. Science is better understood as an iterative mechanism—almost algorithmic in nature—that produces, tests, refines, and occasionally rejects knowledge. This is a central idea, because in essence, a data scientist does exactly that: applies the scientific method as a repeatable process of inquiry. We will return to this point shortly when discussing the *Hypothetico-Deductive Method*.

With this foundation in place, it becomes easier to see why Data Science can be viewed as a generalization of Statistics. This is not a trivial claim. Statistics is undoubtedly a science specifically, a formal science, alongside logic and computer science. Formal sciences study abstract systems through reasoning rather than direct experimentation, distinguishing them from natural, social, and applied sciences. Yet classical statistics has been historically grounded in *inductive inference*: reasoning from the particular to the general. Induction allows us to infer broader conclusions from observed cases, but it carries a fundamental limitation: inductive reasoning can *support* a conclusion, but never *prove* it. This problem, famously articulated by David Hume in the 18th century and known as the *problem of induction*, arises from the logical gap between past observations and future expectations.

Bertrand Russell illustrated this issue through the well-known “inductivist turkey”. A turkey observes that every morning at 9 a.m. the farmer brings food. After hundreds of consistent observations—rainy days, sunny days, warm and cold alike—the turkey confidently concludes that it will always be fed at 9 a.m. Yet on Christmas Eve, instead of being fed, it is slaughtered. A single contradictory event collapses a conclusion seemingly supported by

overwhelming evidence. The lesson is clear: no amount of past observations can logically ensure the truth of a universal claim about the future. Karl Popper pushed this idea further: induction can never confirm a theory with certainty, whereas a single counterexample can refute it. The turkey’s mistake—rooted in the “turkey illusion”—was assuming regularity without understanding the underlying mechanism (the farmer’s intention).

None of this implies that induction is a “bad” method. It remains essential as the generator of hypotheses and conjectures. As Klimovsky notes [16], induction may be seen as a probabilistic justification: it evaluates how strongly observations support a hypothesis, rather than certifying its truth. Put simply, science advances by combining induction (to propose hypotheses) and deduction (to test them). Deduction enables us to assess whether predictions derived from hypotheses correspond to reality. Verification, however, does not grant absolute truth; it merely confirms that a hypothesis has survived a test under certain conditions.

This tension lies at the heart of Leo Breiman’s critique of classical statistics. While statistics provided the mathematical language to quantify uncertainty and draw inferences about populations from limited data, it remained fundamentally bound to inductive reasoning: no matter how large the sample or how robust the model, a future observation could always contradict it. Machine Learning transformed this landscape by not only inferring patterns from data but also validating predictive performance empirically. A model is no longer judged solely by how well it explains past data, but by how accurately it predicts *unseen* data. This represents a qualitative shift that operationalizes the hypothetico-deductive paradigm: generating hypotheses (models) and testing them empirically against new observations in a reproducible and quantifiable way.

To illustrate this in a Data Science context, consider a credit-scoring model. Classical statistics may estimate relationships between customer features and default risk through inductive inference on historical samples. A machine learning model, however, must *demonstrate its predictive power* on new customers those the model has never encountered. If the predictions fail, the model is rejected or refined. This is falsification in action, applied to data-driven decision-making.

In this sense, Data Science can be expressed as **Inference + Prediction**, or equivalently, as **Statistics + Machine Learning**. It unites the inductive strength of statistics with the deductive rigor of model testing, fulfilling Breiman’s vision of a discipline that bridges explanation and prediction a modern science of learning from data. The next section will examine this duality through the lens of

the *Hypothetico-Deductive Method*, the framework that ultimately solidifies the scientific character of Data Science.

14.3 The Scientific Mindset of a Data Scientist

At the core of scientific inquiry lies a logical structure that transcends individual disciplines: the **Hypothetico-Deductive Method**. It provides the conceptual backbone of the scientific method, linking theoretical conjecture with empirical observation. The process begins with a hypothesis—a tentative model proposed to explain observed behavior or predict its future course. From this hypothesis, one deduces empirically testable consequences. These predictions are then confronted with real data through experimentation or systematic observation. When evidence contradicts the hypothesis, it must be revised or discarded; when the evidence supports it, the hypothesis gains provisional credibility, though never definitive truth.

This dynamic interplay between theoretical formulation, deduction, and empirical scrutiny defines the essence of scientific work. What grants scientific legitimacy to a model is not its mathematical elegance or computational sophistication, but its **capacity to generate falsifiable predictions**. The scientific method is therefore inherently self-correcting: each cycle of hypothesizing, testing, and refining moves knowledge closer to truth while acknowledging that truth remains an ever-distant horizon.

Traditional inferential statistics implements this logic only partially, primarily through hypothesis testing. A null hypothesis is proposed, its probabilistic implications are derived, and sample data are used to evaluate whether they justify its rejection. Yet this remains predominantly an inductive process, generalizing from finite observations into probabilistic claims about broader populations. Moreover, statistical validation is typically static once the data are gathered, the hypothesis is evaluated solely in that context, rather than through iterative confrontation with new evidence.

Machine learning, by contrast, operationalizes the deductive stage of scientific reasoning through **predictive validation**. Each model can be viewed as a hypothesis about the functional relationship between predictors and outcomes. Training a model corresponds to formulating the hypothesis; evaluating it on unseen data corresponds to testing its empirical consequences. Performance metrics

quantify how well the model withstands empirical scrutiny, and failures become informative, guiding the reformulation or replacement of the underlying hypothesis.

This perspective turns Data Science into a **practical enactment of the Hypothetico-Deductive Method**. It combines inductive discovery of structure within data with deductive testing of the predictive consequences of that structure. More importantly, it enables Data Science to progress from mere correlation to **genuine explanation**, seeking models that capture mechanisms rather than patterns alone.

Consider the problem of **customer churn** in retail. A predictive model might accurately identify customers at risk of leaving based solely on behavioral correlations a purely inductive exercise. But if the goal is to determine whether a discount campaign will *reduce* churn, prediction alone is insufficient. The data scientist must investigate—and, when possible, formalize—**causal models** capturing how interventions alter outcomes. Prediction answers *who* is likely to churn; causality answers *why* and *what we can do about it*.

Under the Hypothetico-Deductive paradigm, this iterative cycle can be summarized visually, as shown in Figure 14.1, and is typically composed of the following stages:

- **Systematic observation** of a phenomenon
- **Formulation of hypotheses** or models to explain observed patterns
- **Deduction of testable predictions** grounded in those hypotheses
- **Empirical evaluation** of predictions via new data
- **Revision or replacement** of hypotheses based on the evidence collected

Viewed from this angle, the role of a data scientist extends far beyond executing algorithms or statistical routines. It is the **iterative application of the scientific method** across diverse business contexts generating hypotheses, building predictive and causal models, confronting them with evidence, and allowing data to arbitrate which explanations survive. Data Science thus emerges not merely as a technical discipline, but as the **contemporary expression of scientific thinking** applied to data-rich environments, where distinguishing correlation from causation and prediction from explanation defines true professional excellence.

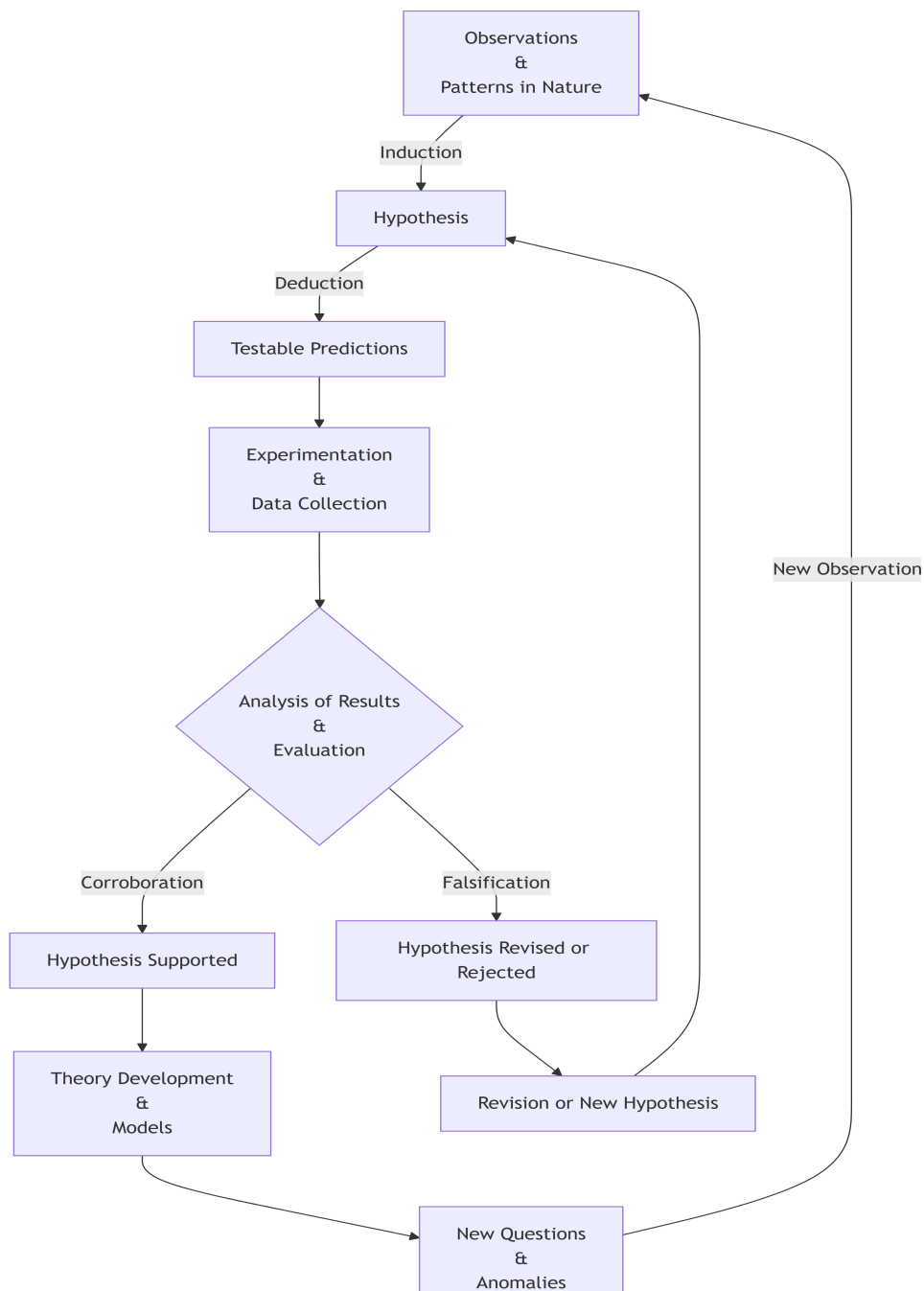


Figure 14.1: The hypothetico-deductive method as an iterative loop: observations motivate hypotheses (induction); hypotheses yield testable predictions (deduction); empirical results either support or challenge the hypothesis, leading to theory formation or revision and ultimately to new questions that restart the process.

14.4 The Fundamental Constituents of Data Science

Despite its rapid expansion and increasing visibility, Data Science is often portrayed as a discipline defined by an ever-growing collection of tools, frameworks, and technological trends. New libraries, platforms, and modeling techniques appear constantly, creating the impression that the field evolves primarily through technological novelty. This perception, however, can be misleading. While modern practice indeed relies on a rich computational ecosystem, the scientific core of Data Science is not defined by software stacks or fashionable methodologies. Rather, it rests on a far more compact and principled foundation: **probabilistic reasoning and statistical inference**.

From a first-principles perspective, Data Science is best understood as a systematic framework for reasoning under uncertainty using data. Probability provides the language to model randomness and variability, while Statistics supplies the methodological machinery for learning from data and evaluating the reliability of the conclusions we draw from it. Together, they form the conceptual backbone that allows data-driven reasoning to remain coherent, interpretable, and scientifically grounded.

Computation plays a crucial—but often misunderstood—role in this framework. The term *computation* derives from the Latin *computatio*, formed by the prefix *com* (“together” or “with”) and *putatio*, related to *putare*, meaning “to calculate” or “to reckon.” In its original sense, the word conveys the idea of **calculating together**. In this light, computation can be understood not as an autonomous source of knowledge, but as a process through which machines assist human reasoning by carrying out calculations at scales and speeds beyond our natural capacity. Algorithms, programming languages, and computational environments therefore act as instruments that enable probabilistic models to be simulated, statistical procedures to be implemented, and empirical hypotheses to be tested. Computation does not define Data Science; it provides the **operational medium** through which probabilistic and statistical reasoning becomes executable in practice.

14.5 Data Science as a Strategic Value Engine

At this point, it is crucial to emphasize a key principle that often gets overlooked: **Data Science is not a goal in itself**. No matter how sophisticated the methodolo-

gies or how advanced the machine learning models, they are worthless if they do not generate real and measurable impact. The true purpose of Data Science is to inform and improve strategic decisions—transforming data into outcomes that advance organizational goals.

Data-driven decision making empowers organizations to optimize operations, personalize customer experiences, identify risks proactively, and maintain a competitive advantage. As Kozyrkov highlights, outstanding data analysts ensure that their work is focused on solving the **right business problems**, not simply producing technically impressive outputs [17].

This underscores a critical truth: technical excellence without business alignment leads to solutions that may appear successful internally but **fail to influence decisions or create value**. The impact of a model is not measured solely by statistical accuracy or computational ingenuity, but by the extent to which it **changes what the organization does**.

From a decision-theoretic perspective, integrating Data Science into strategy enables measurable improvements not only in profitability or utility, but also in **decision velocity**. Automated analytical pipelines allow organizations to deploy models directly into operational environments continuously monitoring performance, adapting to new data, and discarding approaches that no longer deliver value. This creates a self-correcting loop where models are evaluated and refined over time, accelerating the pathway from insight to action.

Consider, for example, the case of **customer churn** in retail. Without Data Science, retention strategies typically rely on intuition or broad marketing campaigns, often costly and ineffective. In contrast, a churn prediction model can **identify customers most at risk**, enabling targeted interventions such as personalized offers or loyalty programs. If the strategy does not reduce churn among the targeted segment, performance monitoring mechanisms reveal the failure, prompting refinement or alternative plans. Here, data does not merely describe behavior—it actively shapes business outcomes.

Viewed through this lens, Data Science becomes a **strategic value engine**: a discipline that leverages scientific reasoning, computational tools, and domain expertise to drive decisions that matter. This synthesis—model-driven insight aligned with real-world objectives—ensures that Data Science fulfills its true mission: transforming information into impact.

14.6 Why the Scientific Method Still Matters

The strategic impact of Data Science ultimately stems from something deeper than predictive models or analytical pipelines. Its real strength lies in the intellectual framework that guides how data is interpreted, hypotheses are formulated, and decisions are evaluated. The ability of Data Science to generate meaningful impact in organizations is therefore inseparable from the scientific principles that structure its practice.

In an era defined by rapid technological change and the constant emergence of new tools, it is tempting to define Data Science by its most visible artifacts: programming languages, machine learning libraries, cloud platforms, and ever-evolving frameworks. Yet these elements, while powerful, are ultimately transient. Technologies evolve, tools are replaced, and methodologies shift as the field continues to mature.

What endures is something far more fundamental: the scientific mindset that underlies the discipline. At its core, Data Science is not about algorithms but about **reasoning under uncertainty**, formulating hypotheses, confronting them with data, and refining our understanding of complex phenomena through systematic evidence. Probability provides the language for uncertainty, statistics supplies the methods for inference, and computation serves as the medium through which these ideas become operational.

When practiced in this way, Data Science becomes more than a collection of analytical techniques—it becomes a modern expression of the scientific method applied to data-rich environments. Its true power lies not in automation or predictive accuracy alone, but in its capacity to transform data into knowledge and knowledge into better decisions. In the end, the enduring value of Data Science does not come from the sophistication of its tools, but from the rigor of the thinking that guides their use.

The original version is available on Medium [here](#).

Chapter 15

Conformal Geometry and Electrostatics in Three Dimensions

15.1 Laplace's equation

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (15.1)$$

15.2 Conformal Mappings

15.3 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

15.4 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

15.5 Data and Methodology

This section describes the data sources, preprocessing steps, simulation procedures, and experimental design used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup

The goal is to provide enough detail to understand how the analysis was conducted while maintaining a clear and readable narrative.

15.6 Implementation

This section describes the computational implementation of the project.

Typical topics may include:

- Libraries and tools used
- Workflow organization
- Numerical methods and algorithms
- Model training procedures
- Pipeline structure
- Computational considerations

Implementation details complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts may be included when appropriate.

15.7 Results and Evaluation

This section presents the main outcomes of the project and evaluates model performance.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results

The emphasis is placed not only on performance, but also on understanding model behavior and extracting meaningful insights from the results.

15.8 Discussion and Limitations

This section discusses the implications, assumptions, limitations, and possible extensions of the work.

Typical topics may include:

- Model assumptions and their consequences
- Practical limitations
- Sources of uncertainty
- Robustness and generalization

- Trade-offs associated with the methodology
- Potential future improvements

This section is particularly important because it reflects critical thinking, technical maturity, and awareness of the limitations of quantitative models.

15.9 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

As shown in Figure 15.1, ...

Suppose we have a region R in xyz space in which a scalar field ϕ satisfies Laplace's equation Equation 15.1. We want to determine what conditions a transformation $f : R \rightarrow R'$ must satisfy such that Laplace's equation remains invariant under this transformation. For simplicity, let's consider the simplest case in $\mathbb{R}^2$.

Applying the chain rule we have that:

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= \frac{\partial \phi}{\partial x'} \frac{\partial x'}{\partial x} + \frac{\partial \phi}{\partial y'} \frac{\partial y'}{\partial x} \\ \frac{\partial \phi}{\partial y} &= \frac{\partial \phi}{\partial x'} \frac{\partial x'}{\partial y} + \frac{\partial \phi}{\partial y'} \frac{\partial y'}{\partial y}\end{aligned}\tag{15.2}$$

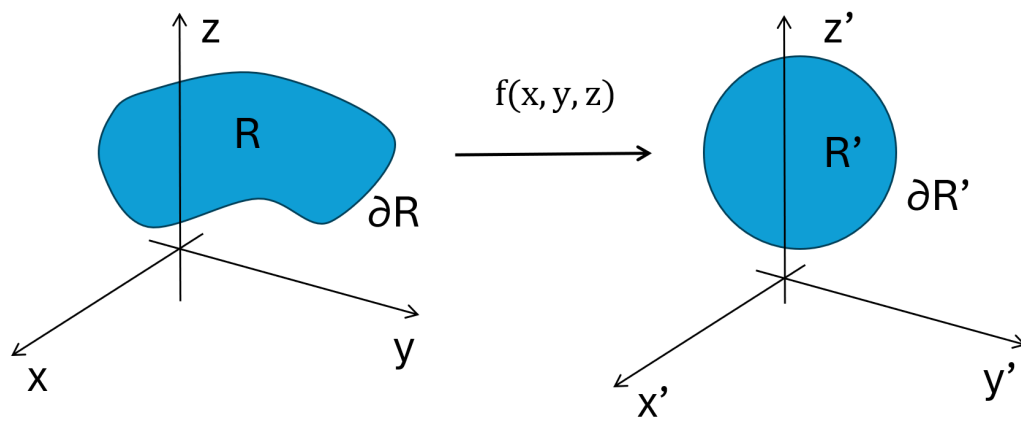


Figure 15.1: Schematic representation of a conformal transformation $f(x, y, z)$ mapping a region R with boundary ∂R in the (x, y, z) coordinate system into a transformed region R' with boundary $\partial R'$ in the (x', y', z') coordinate system.

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